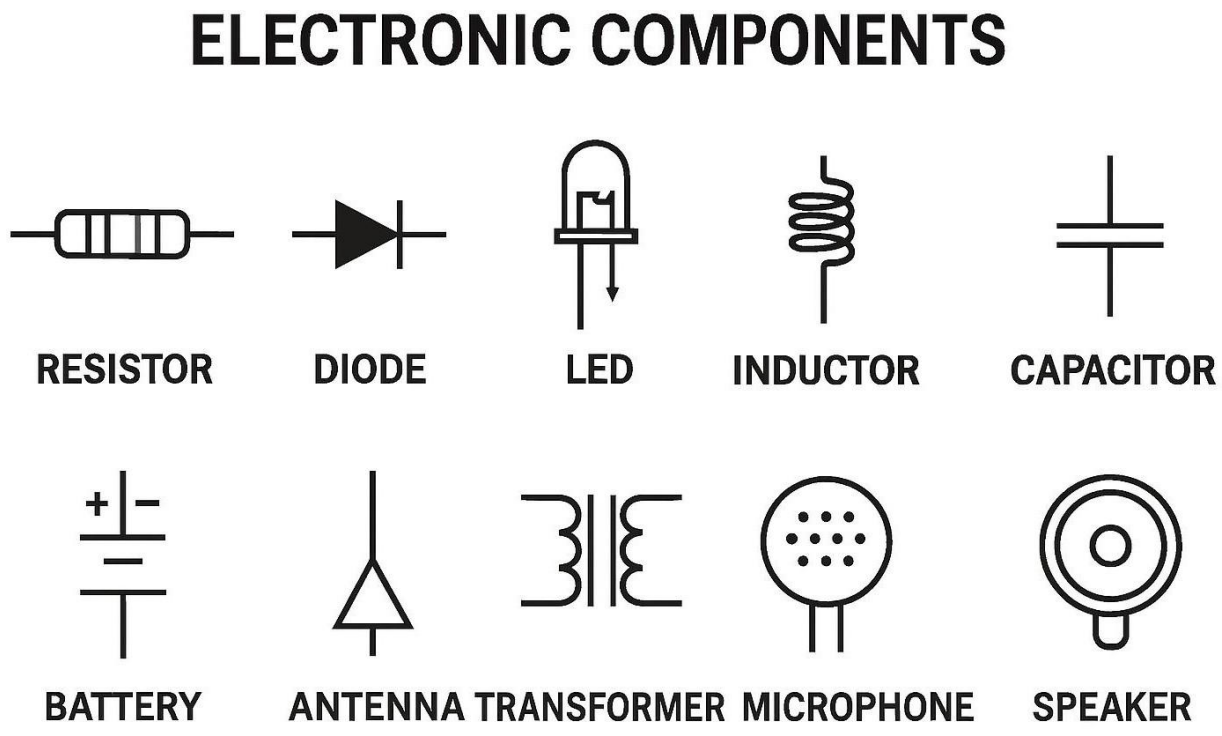
Electronic Components

An electronic circuit consists of different types of basic electronic components comprised of both active components like transistor, diode, IC and passive components like resistor, capacitors, inductors (coil) etc.

These basic electronic components are need for any electronic circuit to function. A Circuit or PCB is on no use without these components. The components can be through-hole with 2 to 3 terminals or they can be SMD Components such as Semiconductor Devices or Integrated Circuits (ICs).

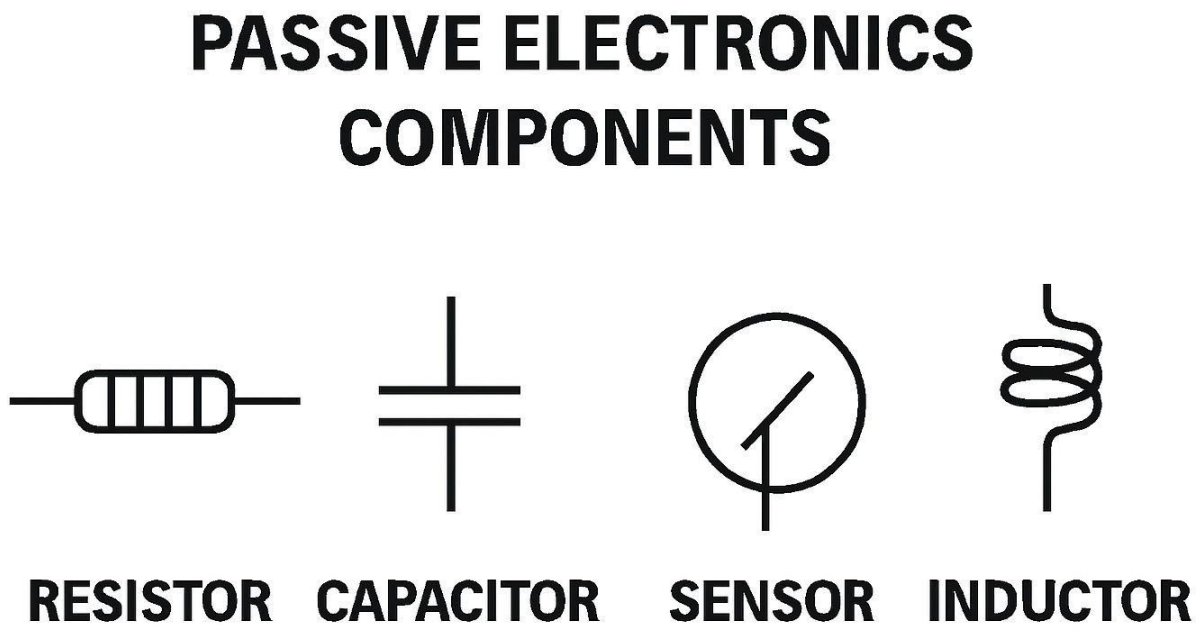
Basic Electronics Components

An electronic circuit or PCB consists of different types of basic electronic components comprised of both Active and Passive Electronic Components like transistor, diode, IC, resistor, capacitors, inductors (coil) etc.



Passive Electronic Components

Passive electronic components are those that can't control electric current by means of another electrical signal. Examples of basic passive electronic components are capacitors, resistors, inductors, transformers etc. These devices can be either Thru-Hole of SMD Components.



Resistor

A Resistor is an electrical device that resists the flow of electrical current. It is used to control or impede the flow of electric current in an electric circuit by providing resistance, thereby developing a drop in voltage across the device.

Consider current flow in terms of water flow. A good electrical conductor is like big water pipe. Water mains and fire hoses have their uses, but you do not want to take a drink from one. Rather, we use small pipes, valves, and other devices to limit water flow to practical levels.

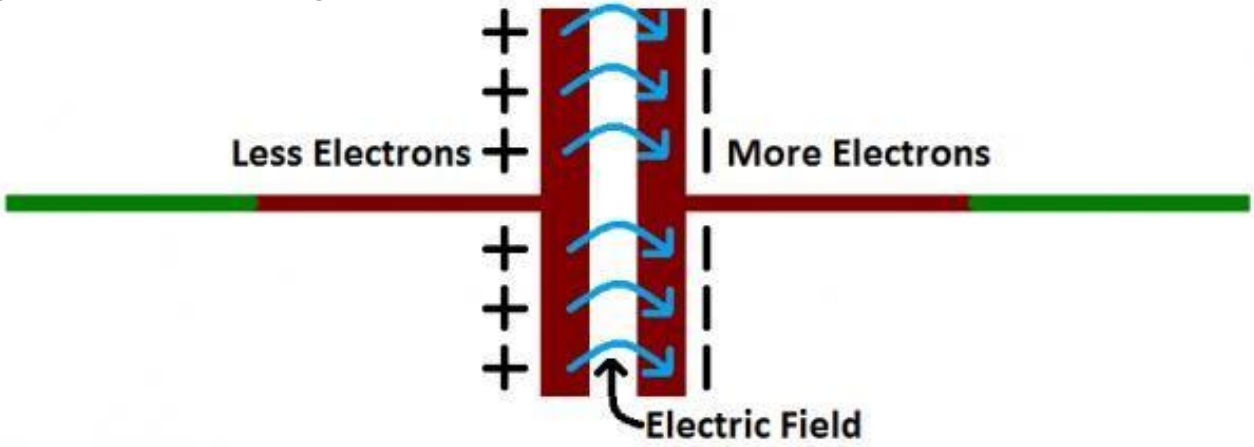
The value of a resistor is measured in ohms and represented by the Greek letter capital omega (**Ω**).

Capacitor

A capacitor is a device that stores electricity for a short time. It has two metal plates separated by an insulator called a dielectric.

When electricity flows into the capacitor, electrons gather on one plate, making it negatively charged. These electrons push away electrons on the other plate, making that plate positively charged. This creates a stored electric charge between the plates.

The capacitor can later release this stored energy back into the circuit. It is useful for smoothing voltage, storing energy temporarily, or letting AC pass while blocking DC.

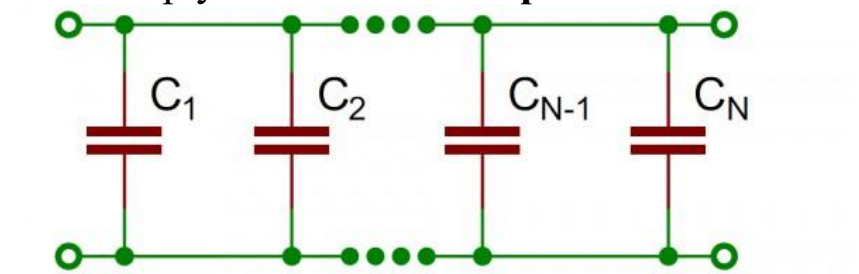


There are several types of capacitors for different applications and functions but Most Common on them are – Ceramic Capacitors, Electrolytic Capacitor, Film Capacitor and Variable Capacitor.

There are two capacitor symbols generally used in electronics. One symbol is for polarized capacitors, and the other is for non-polarized capacitors. The SI unit of capacitance is farad (Symbol: **F**).

Capacitors in Parallel

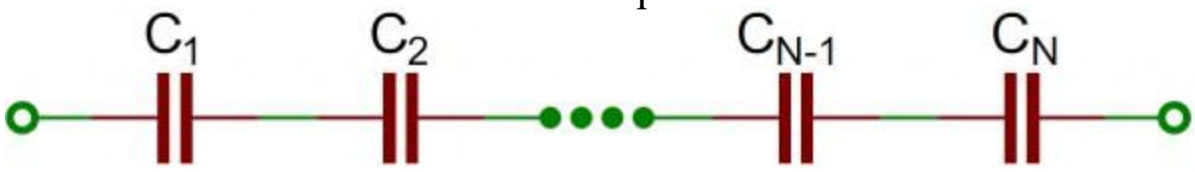
When capacitors are placed in parallel with one another the total capacitance is simply the **sum of all capacitances**.



$$C_{Tot} = C_1 + C_2 + \ldots + C_{N-1} + C_N$$

Capacitors in Series

Much like resistors are a pain to add in parallel, capacitors get funky when placed in *series*. The total capacitance of *N* capacitors in series is the inverse of the sum of all inverse capacitances.



$$\frac{1}{C_{Tot}} = \frac{1}{C_1} + \frac{1}{C_2} + \ldots + \frac{1}{C_{N-1}} + \frac{1}{C_N}$$

Inductor

An inductor is a passive electrical device that introduces inductance into a electric circuit. It is basically a coil of wire with many windings, often wound around a core made of a magnetic material, like iron. Simplest form of an inductor is made up of a coil of wire.

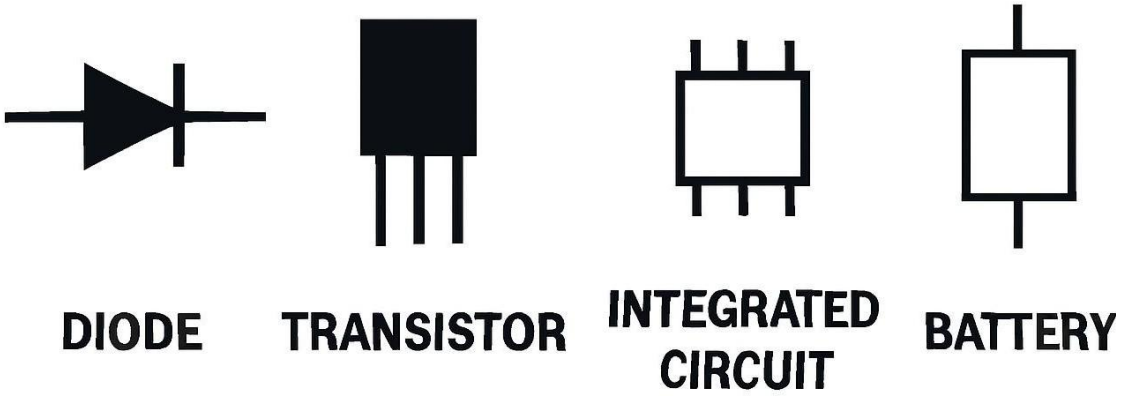
The SI Unit of Inductance is henry (H) named after the American scientist Joseph Henry.

Inductors are categorized into different types but most common of them are – Air Cored Inductor, Iron Cored Inductor, Ferrite Cored Inductor, Iron Power Inductor, Multilayer Ceramic Inductors, Film Inductors, Variable Inductors.

Active Electronic Components

Active electronic components are those that can control the flow of electricity. Different Types of Printed Circuit Boards have at least one active component. Some examples of active electronic components are Diode, Transistor, Vacuum Tubes, Silicon-Controlled Rectifiers (SCRs), ICs.

ACTIVE ELECTRONIC COMPONENTS



Diode

A diode is a device made with semiconductor material that allow flow of electricity in one direction. Most diodes have a painted line on one end showing the direction or flow. It has two terminals – anode and cathode. The negative side is normally white.

They are most commonly used to convert AC to DC, because they pass the positive (+) part of the wave, and block the negative (–) part of the AC signal, or, if they are reversed, they pass only the negative part and not the positive part.

Some of the most commonly used diodes are – Zener Diode, PN Diode, LED (*Light Emitting Diode*), and Photo Diode.

Transistor

A transistor is a three-terminal semiconductor device. It is generally used as a switching device or as an amplifier. It is a combination of two or more Diodes. Transistors are of two types – Bipolar Junction Transistor (*BJT*) and Field Effect Transistor (*FET*). They can be PNP (*Positive-Negative-Positive*) and NPN (*Negative-Positive-Positive*) transistors.

Integrated Circuit (IC)

Integrated Circuit or IC is an SMD Electronic Component made up of combination of several transistors, diode, resistor, capacitors in a tiny semiconductor chip. Integrated Circuit Electronic Components or ICs are of small size and very light weight but produce excellent results at low power.

Components used inside an IC are not visible from outside. These are built in a semiconductor chip and cannot be removed. Capacitors of more than 30 PF value are not used in an IC nor can inductors and transformers be mounted inside the chip of an IC. Thousands and millions of bi-polar transistors and CMOS (*Complementary Metal Oxide Semiconductor*) FET are used in making an IC.

Based on external structure, ICs or SMD can be classified as follows:

- Single in-line pin package (SIPP).
- Dual in-line pin package (DIPP).
- Quard pin package (QPP).
- Pin Grid Array Package (PGA).
- Ball Grid Array Package (BGA).
- Leadless Chip Carrier (LCC) Package.

Battery

A battery is a device with a collection of one or more cells that forces flow of electrons in a circuit through electrochemical reaction. It actually transforms stored chemical energy directly into electrical energy. All types of batteries have: an anode (Negative *Terminal*), a cathode (Positive *Terminal*), and some chemical called electrolyte for chemical reaction.

References:

https://www.allaboutcircuits.com/textbook/direct-current/chpt-13/electric-fields-capacitance/